BEFORE THE
STATE OF NEW YORK
PUBLIC SERVICE COMMISSION

Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations of Consolidated Edison Company of New York, Inc. for Gas Service.

Case 25-G-0073

June 24, 2025

Rebuttal Testimony of:

Gas Forecasting Panel

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CONSOLIDATED EDISON COMPANY OF NEW YORK, INC.

GAS FORECASTING PANEL
REBUTTAL TESTIMONY

I. INTRODUCTION

Q. Please state your names.

A. Kathleen Trischitta, Andrew Nichols, and Johan Tolosa.

Q. Have you previously submitted testimony in this proceeding?

A. Yes. We previously submitted direct testimony and update testimony on behalf of Consolidated Edison Company of New York, Inc. ("Con Edison" or "Company") as the Gas Forecasting Panel ("Panel").

Q. Is the Panel rebutting any testimony from Staff and/or intervenors?

A. Yes, Staff filed answering testimony in this proceeding on the Panel's direct and update testimonies to which the Panel will rebut. The issues raised by the Staff Forecasting Panel to which we will rebut include:

- whether the Company's weather and usage data is adequate to produce a reliable regression analysis and accurately estimates weather impacts that deviate from normal;
- whether a 10- or 30-year average normal weather assumption is appropriate;
- whether the inclusion of a climate trend adjustment in the Panel's gas delivery volume forecast is appropriate;
- whether the Staff Forecasting Panel's post-model

adjustments for energy efficiency are appropriate.

Q. Do you have any exhibits that accompany this testimony?

A. Yes, we are presenting two exhibits, Exhibit __ (GFP-9) and Exhibit __ (GFP-10).

Q. Were these two exhibits prepared under the Panel's direction and supervision?

A. Yes. We describe each of these exhibits below in our testimony.

II. REBUTTAL

Q. What is the Staff Forecasting Panel's recommended gas volume forecast for Rate Year 1?

A. In testimony, Staff recommends (at 11:17-11:20) an increase of 3,010 thousand dekatherms, or MDt, to the Company's gas volume forecast of 165,630 MDt. In response to Company IR-83 (see Exhibit AP-8) Staff corrected its forecasted gas delivery volume to be 165,823 MDt, resulting in a difference of 193 MDt (or 0.1%) relative to the Company's forecast.

Q. The Staff Forecasting Panel contends (at 29:6-29:12) that using one year's worth of weather and usage data does not give an accurate estimate of weather impacts that deviate from normal, and the nine data points associated with the

heating season months of the base year is not adequate for a reasonable regression analysis. Do you agree?

A. We do not agree. The regression used by the Company is designed specifically for short-term normalization of a single year's usage — not for long-term forecasting. Using recent data ensures alignment with current customer behavior and avoids incorporating outdated or structurally inconsistent patterns, which is especially important in a post-pandemic environment. While the model uses nine months of heating season data, it is supplemented with targeted adjustments for customer growth, energy efficiency, and electrification. These enhancements improve forecast accuracy and ensure the methodology remains responsive to near-term system dynamics. Furthermore, the approach of using the most recent heating season months as used by the Gas Forecasting Panel is consistent with the approach used by the Company in its annual Weather Normalization Adjustment filing with the Public Service Commission each August. This approach is outlined in Section IX. Special Adjustments 1. Weather Normalization Adjustment as found in the Company's Gas Tariff.

Q. The Staff Forecasting Panel further contends (at 30:14-33:6) that 2024 may not represent a typical year. Is this correct?

A. The Company agrees, as it stated in initial testimony. However, the Company believes that its bottom-up approach in forecasting volumes is better suited to address atypical years. In such an approach, the Company is able to make adjustments to deviations from a typical year whereas Staff's econometric forecast does not demonstrate the same capabilities. Such adjustments made by the Company and not accounted for by Staff include reallocating volumes across months, and sometimes across years, for known billing discrepancies.

Q. The Staff Forecasting Panel similarly asserts (at 31:2-31:5) that the Panel's adjustments to its volume forecast (which included customer growth, electrification, and new oil-to-gas customers) were improperly based on only one year of per-customer usage data, because it does not account for the impact of recent swings in the economy on energy usage. Why is it appropriate to use one year of per-customer usage data to derive volume forecast adjustments as the Panel did in its direct testimony?

A. This approach is appropriate because of how the forecast is produced. The forecast is built upon a 12-month base period with a known average use per customer. The forecast then layers forecast modifiers, such as energy efficiency, which ultimately results in an updated average use per customer for each year and each month of each service classification. In addition, the Company incorporates economic data into its methodology, contrary to the Staff Panel's assertion. Specifically, the Company uses New York City Gross Domestic Product (GDP) and private non-manufacturing employment (PNEMP) as key economic indicators in its customer models. These variables are critical for capturing macroeconomic trends that influence gas demand across customer classes. Additionally, the Company uses these economic inputs to inform adjustments related to customer growth, oil-to-gas conversions, and electrification impacts. These data points were provided in responses to Staff Information Requests, including DPS-271 (SSFP-1, page 104), and are integral to the Company's hybrid approach, which blends empirical economic data with operational insights to produce a forecast that is both statistically grounded and operationally relevant.

Q. The Staff Forecasting Panel disagrees (at 29:15-29:17) with the Panel's use of a 30-year average normal weather assumption and instead recommends using a 10-year average to normalize gas delivery volumes. Why is the Company using a 30-year average normal weather assumption?

A. The Company is using a 30-year average to normalize gas delivery volumes as a result of Case 16-G-006. The testimony describing Staff Gas Rate Panel's recommended modifications to the Company's forecast states: "Based on the recommendation of the Staff Gas Supply and Reliability Panel, we used 30-years of data to determine normal HDDs compared to the Company's method of using 10-years of data." The switch from using a 10-year average to a 30-year average is also noted on Page 69 of the Joint Proposal describing the Company's Weather Normalization Adjustment.

Q. Unlike the Panel's forecasts, the Staff Forecasting Panel did not include an adjustment for climate change. The Staff Forecasting Panel contends (at 18:15-19:3) that including such an adjustment requires a study and climate trends are long-term and thus may not be applicable to a short-term forecast that the Panel uses for forecasting.

Please explain why it is appropriate to include a climate trend adjustment in the Company's forecasts.

A. Given that this topic is common to both electric and gas, it is more efficient to address these issues in one place (i.e., in the rebuttal testimony of the Company's Electric Forecasting Panel) than to repeat these same arguments here. The Gas Forecasting Panel hereby adopts the rebuttal testimony of the Electric Forecasting Panel, as it relates to the appropriateness of a climate trend adjustment as its own.

Q. The Staff Forecasting Panel contends (at 29:18-29:22) that the 2009 Central Hudson Rate Order determined a 10-year average weather normalization more reasonably reflects anticipated weather trends. Do you agree?

A. No. The Company does not agree with this statement particularly as it relies on a 2009 Commission determination in the Central Hudson rate case. That assessment, now over 16 years old, was made in the context of a different utility, operating in a different geographic region with distinct weather patterns and a substantially different customer base. Applying that determination to Con Edison's current service territory overlooks the unique characteristics of New York City's

climate, infrastructure, and energy usage trends. The Company also notes that Staff's testimony recommending a 30-year average for Con Edison occurred in 2016, after the 2009 case Staff is relying on for precedent.

Q. Does the Panel have any concerns with Staff's energy efficiency adjustments to their forecast models?

A. Yes. The Panel does not agree with Staff's approach to their energy efficiency adjustments. While the Staff Panel adjusts its gas forecast to account for energy efficiency, its methodology relies on extrapolating historical EE trends without incorporating forward-looking information about future program funding, implementation strategies, or customer participation rates. This backward-looking only approach assumes that past energy efficiency performance is a reliable predictor of future outcomes, which may not hold true in the presence of evolving policy mandates, incentive structures, or shifts in customer behavior. Statistically, this introduces a risk of model misspecification due to omitted variable bias—where key explanatory factors, such as planned EE investments or programmatic changes, are excluded from the adjustment. In contrast, the Company's approach integrates both historical data and known future programmatic inputs,

allowing for a more complete and policy-aligned estimate of EE impacts. Ignoring these forward-looking elements may understate or misrepresent the true effect of energy efficiency on future gas consumption.

Q. Does the Panel have any additional concerns with Staff's energy efficiency adjustments?

A. Yes. The Panel does not agree with Staff's approach of neglecting the impact of organic energy efficiency. The Company uses internal subject matter expertise to derive a forecast of organic energy efficiency. This forecast is driven by forecasted improvements to gas equipment, caused by gas customers replacing existing equipment with more efficient models as well as building renovations across the service territory. Staff's assumption implies that historical patterns of organic efficiency such as behavioral changes, appliance turnover, and building code improvements will remain static, which is unlikely. These factors evolve over time and can have a compounding effect on usage, even in the absence of formal programs. Moreover, Staff's position is inconsistent with the goals of New York State's Climate Leadership and Community Protection Act, which anticipates less fossil fuel usage. It also overlooks the reality that buildings, businesses,

and homeowners are increasingly replacing gas-fired appliances and improving efficiency independent of utility programs.

Q. What concerns does the Company have with Staff's customer forecasting models?

A. For several customer forecasting models used by Staff, there are issues with using non-stationary data.

Q. Why does the Company feel there are stationarity issues and what does that mean for the models used by Staff?

A. Exhibit __ (GFP-9) provides either an Augmented Dickey-Fuller test and/or a seasonal graph for each of the following variables:

- For the SC1 customer model, the variables dlog(SC1_CUST,0,12), and dlog(POP, 0, 12);
- for the SC11 customer model, dlog(SC11_CUST_ADJ);
- for the SC2 R1 customer model, dlog(SC2R1_CUST); and
- For the SC2 R2 customer model, dlog(SC2R2_CUST_ADJ).

As shown in the exhibit, each of the provided Augmented Dickey-Fuller tests fails to reject the null hypothesis that the respective variable has a unit root at the 5% significance level. As such, each of these tested variables is non-stationary, which violates a basic

requirement of time-series modeling. This renders the results of each of these models, and their associated forecasts, to be of questionable reliability and accuracy.

Q. What do the seasonal graphs show?

A. A seasonal graph displays a series ordered by season. Because the data frequency used for each model is monthly, each series is grouped by month. These graphs exhibit a strong seasonal component in its mean and/or its variance. A data series whose statistical properties such as its mean and/or variance are dependent on the time period it is observed is considered to be non-stationary. This violates a basic requirement of time-series modeling and renders the results of each model, and their associated forecasts, to be of questionable reliability and accuracy.

Q. Does the Company have any concerns with Staff's volume forecast?

A. Yes. The Company has identified issues with non-stationary data.

Q. Which variables used in Staff's volume models have been identified as non-stationary?

A. Please reference the following list:

- For the SC1 Volume model, the variables dlog(SC_1_VOL), and dlog(STAFF_SC1_RES_CUST_FCAST);
- for the SC2R2 Volume model, dlog(SC_2_R2_VOL);
- for the SC11 Volume model, dlog(SC_11_VOL), and MOVAV(D_PRICE_SC11,6); and
- for the SC31 Volume Model, dlog(SC_31_VOL). MOVAV(D_PRICE_SC11,6).

Q. Has the Company created an exhibit to support the claim of non-stationary series used in Staff's volume models?

A. Yes. For each of the above variables, Exhibit __ (GFP-10) shows a seasonal graph.

Q. What do the seasonal graphs show?

A. These graphs exhibit a strong seasonal component in its mean and/or its variance. A data series whose statistical properties such as its mean and/or variance are dependent on the time period it is observed is considered to be non-stationary. This violates a basic requirement of time-series modeling and renders the results of each model, and their associated forecasts, to be of questionable reliability and accuracy.

Q. Do you agree with Staff's removal of the funding for the FTEs associated with maintenance of the Gas Distribution Forecasting Tool (GDFT) application?

A. No. DPS Staff removed the positions based on its assumption that there were overlapping responsibilities in the maintenance of the REV/DER/EEDM application and GDFT application. However, these tools are managed by two separate teams and each serves a completely distinct purpose. The REV/DER/EEDM is a forecasting tool that generates 20-year projections for specific load modifiers such as battery storage, building electrification, electric vehicles, photovoltaics, and distributed generation/combined heat and power on the electric and gas system. It also provides visual summaries that illustrate the impact of the individual load modifiers on network and system peak, volume, and billable demand. This tool is designed as an Azure cloud-based application, composed of JavaScript Single page app for user interaction, and a Python-based analytical engine which generates the outputs that are utilized in the official network and system peak forecasts. The GDFT is a distribution-level firm gas forecasting model designed to project peak demand under design weather conditions with high spatial and temporal

granularity. It incorporates historical meter-level consumption data and uses input variables such as new business, construction, demand response, steam-to-gas conversions, energy efficiency, distributed generation, oil-to-gas conversions, and building electrification. The tool enables 20-year peak day demand forecasts at the district and regulator level for the entirety of the gas distribution system, supporting localized planning and operational reliability. Support for GDFT is important as it allows the Company to incorporate clean energy efforts into our gas distribution system, including identifying areas where non-pipeline alternatives (NPA) are applicable. This is paramount in meeting gas customer growth demands in our territory while transitioning to our clean energy goals.

Q. Did Staff address the Company's proposal for continued funding of the Energy Management Analytics (EMA) application?

A. Yes, Staff recommended to not fund the EMA application with further capital. In addition, Staff disallowed O&M expenses necessary to maintain the NYISO's billing functionality of reviewing the invoices for accuracy and reconcile the payment statements.

Q. Why is it important to continue capital funding of the Energy Management Analytics (EMA) application?

A. The additional EMA capital work enhancements will transform raw operational billing data into strategic decision-making tools for both the company and our customers. The work is to:

- Develop real-time rate change indicators by analyzing energy and gas price trends by service class, enabling immediate visibility into cost fluctuations.
- Create dynamic, service-class-specific infographics that clearly communicate cost rates per hour, significantly improving rate transparency.
- Evaluate localized economic benefits of battery storage deployment across specific network areas to optimize infrastructure investments.

Q. Why did Staff disallow the O&M funding being requested for the EMA capital project for this software?

A. Staff appears to have misinterpreted the request as funding for an additional or partial FTE. With the core software functionalities now established and in Production status, the Company seeks to transition ongoing maintenance to IT or a qualified vendor. This

strategic reallocation will enable the Company's analytical staff to focus on higher-value projects and initiatives that better leverage their specialized expertise. This is an operational sustainment need as the work involves minor code adjustments to accommodate market/billing changes (e.g., new renewable resources like battery storage), and database structural changes. This is a cost-neutral transition of responsibilities, not a new payroll expense.

Q. How will the O&M funds be used if approved?

A. Funds will exclusively cover:

- IT/vendor costs to maintain and update the existing software (e.g., code adjustments for market changes, billing updates).
- Zero FTE payroll allocation: The goal is to free up current analytical staff for higher-priority projects like real-time rate alerts and customer-facing analytics.

Q. What are the risks of not approving EAM O&M request?

A. The risks include:

- Lost opportunity cost: FTEs remain tied to maintenance instead of advancing customer-facing analytics and the project fails to meet professional

standards or achieve widespread adoption across the organization.

- Technical debt: Without vendor/IT support, the software may become outdated as renewable portfolios expand. Vendor or IT supported code updates ensure timely adaptation to new resources (e.g., battery storage) without overburdening internal teams.
- Delayed price alerts: Slower adaptation to market changes reduces our ability to proactively address rate impacts.

As a minimum we recommend the Staff to approve the O&M funding since billing core billing verifications depend on this application with or without capita approval. This O&M request aligns with the Enterprise Data & Analytics strategy, ensuring efficient resource allocation while advancing critical customer protections.

Q: How does EMA infographics and analytics support NY State priorities and operational objectives?

A. The real-time rate indicators developed through this project will directly contribute to:

- Proactive cost management: Early detection of energy price spikes will enable collaborative mitigation

strategies, including optimized hedging and phased rate adjustments.

- Enhanced customer protection: Data-driven insights will allow for bill stabilization over longer periods, preventing or mitigating volatility issues.

These outcomes align with Staff and stakeholder focus on bill stabilization, transparency, and modernized grid management.

Q. Does this conclude the Gas Forecasting Panel's rebuttal testimony?

A. Yes. It does.